PalmSource™ 99

How to Sync with the Enterprise

Ron Kollgaard
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3Com
More connected.

Extend the Enterprise

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Agenda

- A Palm Solution
- Road to the Enterprise
  - The Handheld
  - The Enterprise
  - Synchronization
- Sync to the Enterprise
- What have we learned?



Extend the Enterprise

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Your Enterprise Applications…

- E-mail and PIMs
- Groupware
- Databases
- Web applications
- ERP systems
- Vertical applications



…Everywhere Your Workers Are

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- Inventory control
- Sales teams
- Data gathering
- Vertical applications
- Supply chain control







Extend Enterprise Information

- A solution that will
  - Gather
  - Disseminate
  - Display
- A portable solution

Intelligent Simplicity



Extend Enterprise Information

- A Palm Solution
  - Ease of use
  - Portability
  - Connectivity



Intelligent Simplicity

A Complete Enterprise Solution

- The PDA itself is just the tip of the iceberg





A Complete Enterprise Solution

- The handheld



A Complete Enterprise Solution

- The handheld
- Synchronization software





A Complete Enterprise Solution

- The handheld
- Synchronization software
- Enterprise information







A Complete Enterprise Solution

- Synchronization is the key to success
  - Much of your attention will likely be here







Extend the Enterprise

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- A Palm Solution
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The Road to the Enterprise

- Choose your route wisely before you begin
  - Keep in mind the complete solution





It may not be easy to get back on track

The Road to the Enterprise

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synchronization

conduits

handheld

connectivity

enterprise

data flow



What Will the Palm Do?

- Display information
  - Supply chain management
  - Product details





What Will the Palm Do?

- Display information
  - Supply chain management
  - Product details
- Gather data
  - Inventory warehouse
  - Patient information





What Will the Palm Do?

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- Display information
  - Order status
  - Product details
- Gather data
  - Inventory warehouse
  - Patient information
- Independent computations
  - Configure products
  - Determine rates





What Will the Palm Do?

- Display information
  - Order status
  - Product details
- Gather data
  - Inventory warehouse
  - Patient information
- Independent computations
  - Configure products
  - Determine rates
- All of the above
  - Sales force automation





The Road to the Enterprise

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synchronization

conduits

handheld

connectivity

enterprise

data flow



How Will the Palm Connect?

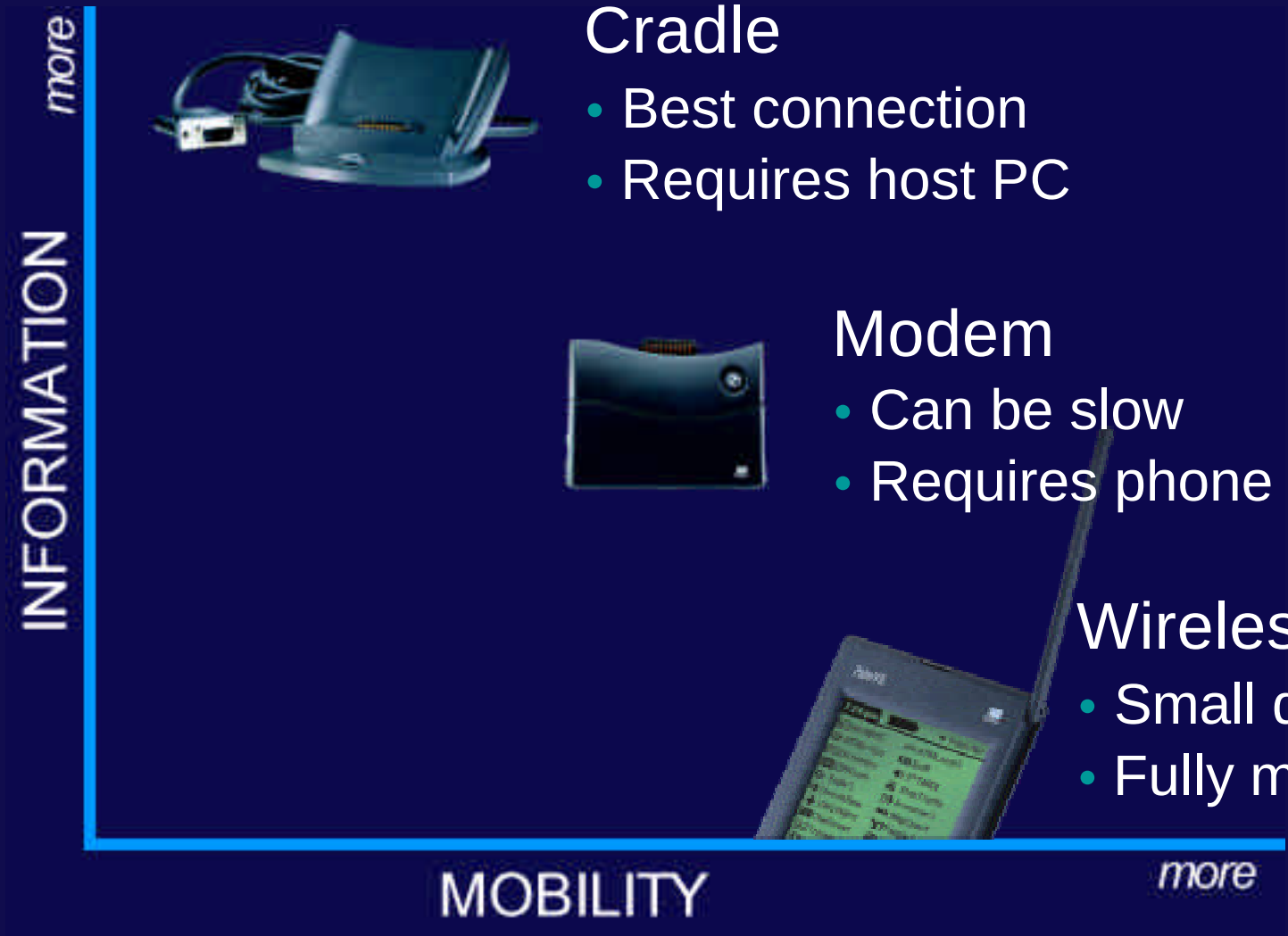
- Cradle
- Modem
- Wireless

via

- Serial Port
- USB
- TCP/IP
- RAS



Connection Tradeoffs



Cradle

- Best connection
- Requires host PC

Modem

- Can be slow
- Requires phone line

Wireless

- Small data packets
- Fully mobile

The Road to the Enterprise

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synchronization

conduits

handheld

connectivity

enterprise

data flow



What Enterprise Data Is Needed?

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- E-mail / PIM
  - Exchange
  - Lotus Notes
  - POP3 / SMTP



What Enterprise Data Is Needed?

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- E-mail / PIM
- Groupware
- Lotus
- Outlook



What Enterprise Data Is Needed?

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- E-mail / PIM
- Groupware
- Databases

- Oracle
- DB2
- Sybase
- SQL Server



What Enterprise Data Is Needed?

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- E-mail / PIM
- Groupware
- Databases
- ERP
- PeopleSoft
- SAP



What Enterprise Data Is Needed?

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- E-mail / PIM
- Groupware
- Databases
- ERP
- Web based

- Corporate web pages
- eCommerce
- News and information



What Enterprise Data Is Needed?

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- E-mail / PIM
- Groupware
- Databases
- ERP
- Web based data
- Vertical applications
- Supply chain
- Customer relations
- Health care
- Sales teams



The Road to the Enterprise

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synchronization

conduits

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How Much Enterprise Data?

- E-mail / PIM
- Groupware
- Databases
- ERP
- Web-based data
- Vertical applications



How Much Enterprise Data?

- Large product catalogs
- Complete inventories
- Map, signature capture, images



A lot



How Much Enterprise Data?

- Price quotes
- Updated contacts
- Ordering information





A little



The Road to the Enterprise

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synchronization

conduits

handheld

connectivity

enterprise

data flow



Server Synchronization Software

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AvantGo	AvantGo Server
GlobalWare	Pylon ServerSync
IBM / Advance Systems	Mobile Connect
Palm Computing	HotSync Server
Puma	IntelliSync Anywhere
Riverbed	ScoutWare
River Run	MobileSuite
Synchrologic	SynchKit
WaveWare	WaveSync

Bridging the Gap

- Connecting the Palm to the Enterprise





Choose the appropriate synchronization solution

Desktop Synchronization

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One-to-one connection





Handheld application

Enterprise data

Server Synchronization



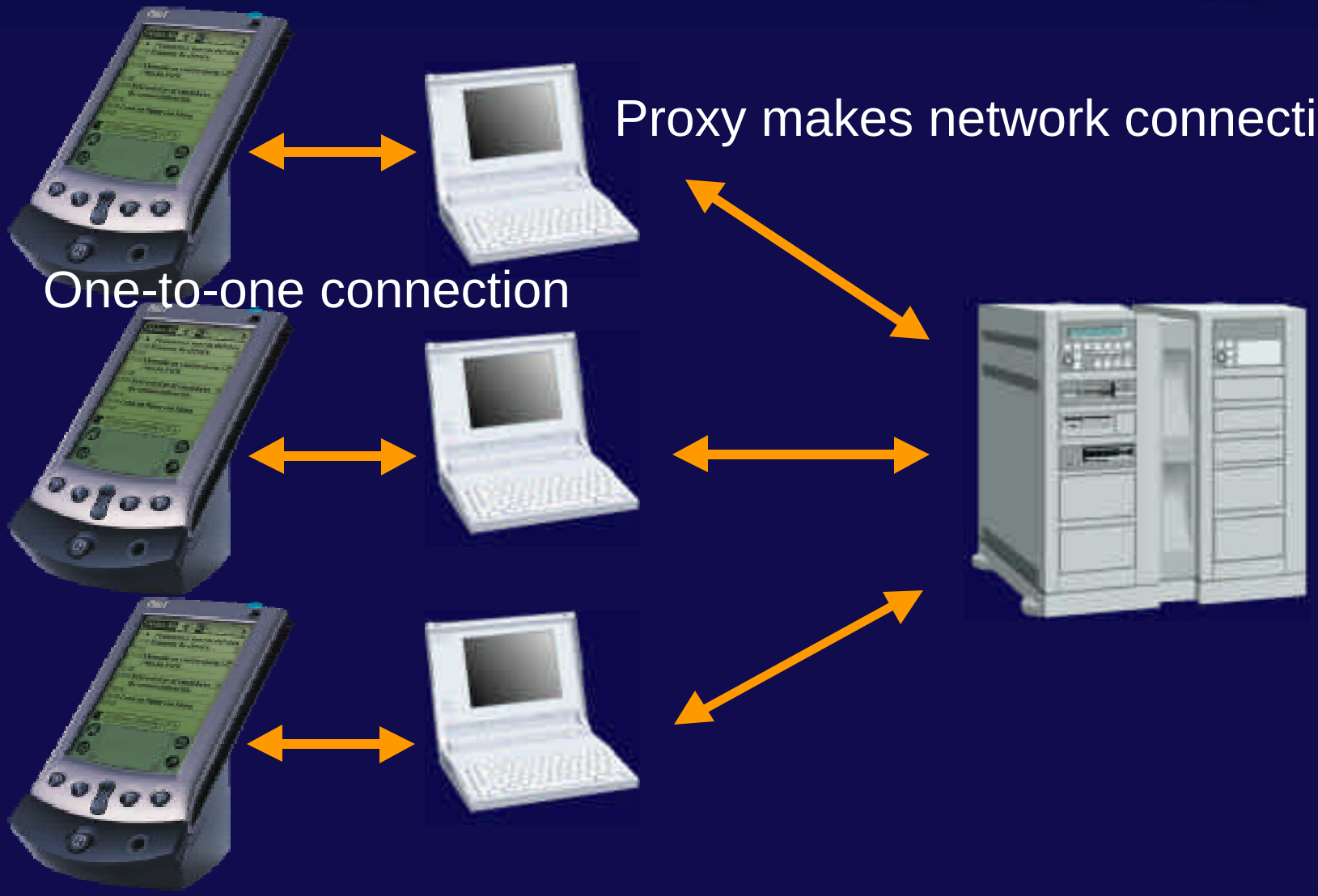
Client application

Many-to-one connection



Proxy–Server Synchronization

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Web–Server Synchronization

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Browser application





Web-based transport protocols



Server Synchronization Solutions

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- Advantages
  - Allows for sharing of enterprise data
  - Centralized control over synchronized data
  - User administration
  - Device administration
  - Can handle security

Proxy–Server Solutions

- Advantages
  - Can be a simpler sync solution for the Palm device
  - Can make device recovery easier

Web–Server Solutions

- Advantages
  - Client app on Palm can be easier to administer
  - Standard transport protocols used
  - Can use existing web infrastructure

The Road to the Enterprise

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synchronization

conduits

handheld

connectivity

enterprise

data flow



Consider Your Conduit

Conduits: the key to synchronization

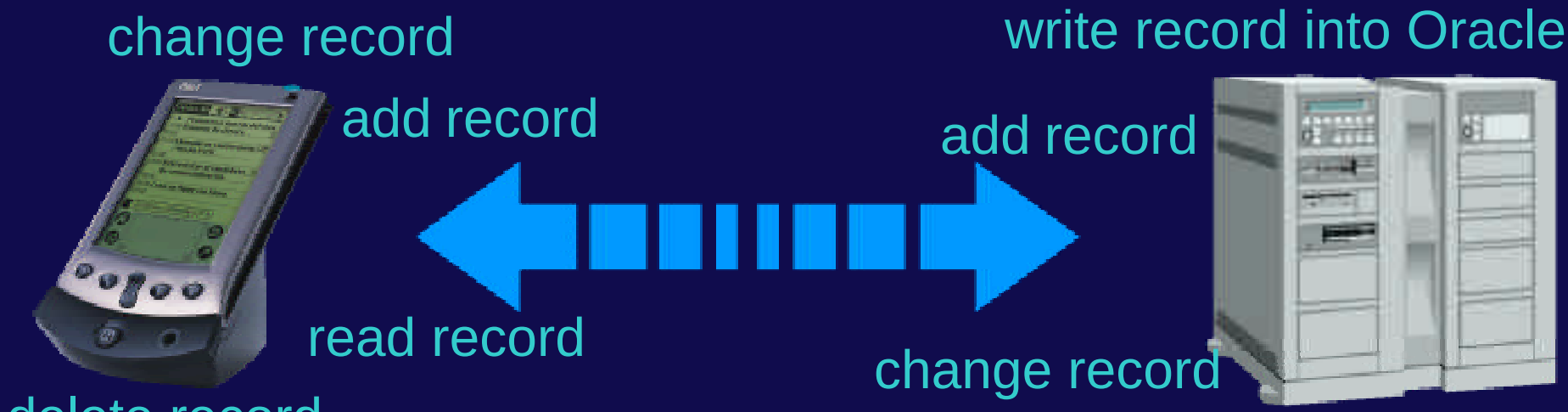


A conduit is the software that controls the synchronization of information between the Palm device and the PC



Consider Your Conduit

Conduits: the key to synchronization



What Will the Conduit Do?

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- Data replication



store name
sales volume



store name
sales volume



What Will the Conduit Do?

- Data replication
- Importing and exporting data



store name
sales volume





What Must the Conduit Do?

- Data replication
- Importing and exporting data
- Decisions



record sales in database

store name
sales volume

Are sales good this quarter?

YES

NO

e-mail HR: fire manager





Conduit Philosophies

- Canned conduits
- Managed conduits
- Programmable conduits



Conduit Philosophies

- Canned conduits
- Managed conduits
- Programmable conduits

Caution:
These definitions are fuzzy



Canned Conduits

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- Install and go
  - Easiest, fastest deployment
  - Product features must match your needs



Canned Conduits

- PylonPro (Globalware Computing)
  - Connecting your Palm devices to Lotus Notes







Lotus Notes Server

Canned Conduits

- Intellisync Anywhere (Puma Technologies)
- Pylon ServerSync (Globalware Computing)

Managed Conduits

- Configure with an administration program
  - Includes scripting (VB Script, JavaScript, etc.)



Managed Conduit

- Sales data in and out of an ODBC database
  - Configure the rules for matching data
  - Add scripting for more complex tasks



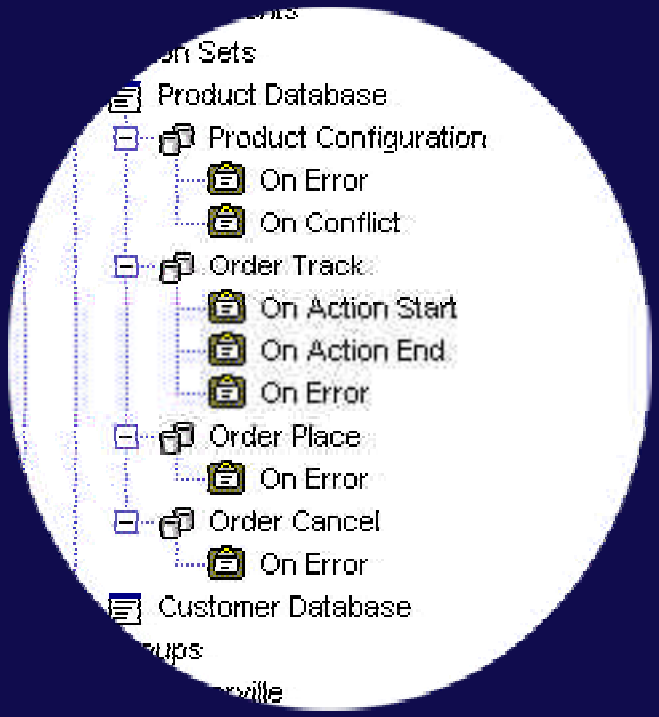


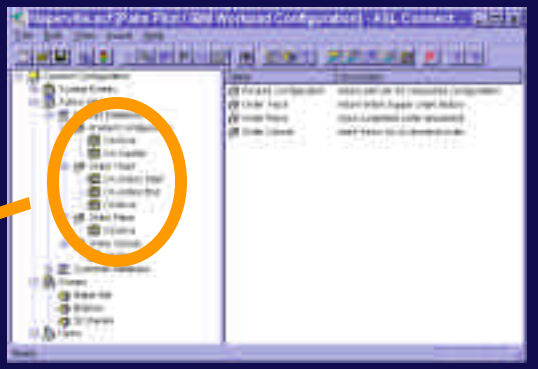


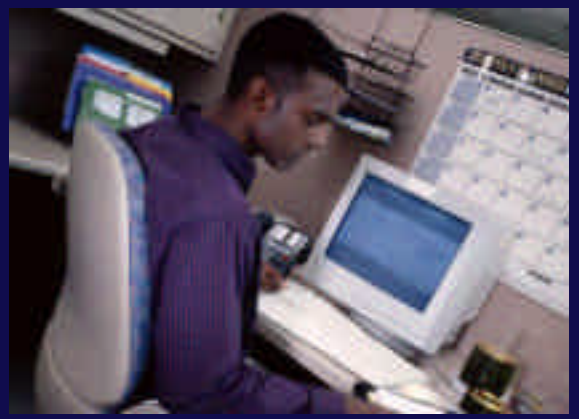
Oracle
Sybase
DB2
SQL Server
Etc.

Managed Conduit

- Mobile Connect (IBM)
  - Hire a “conduit administrator”







Managed Conduits

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- ASL-Connect (Advance Systems)
- MobileSuite (River Run)
- SynchKit (Synchrologic)

Web-based data

- AvantGo Server (AvantGo)

Programmable Conduits

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- Create your own conduit program
  - Greatest flexibility
  - Greatest challenge



Programmable Conduits

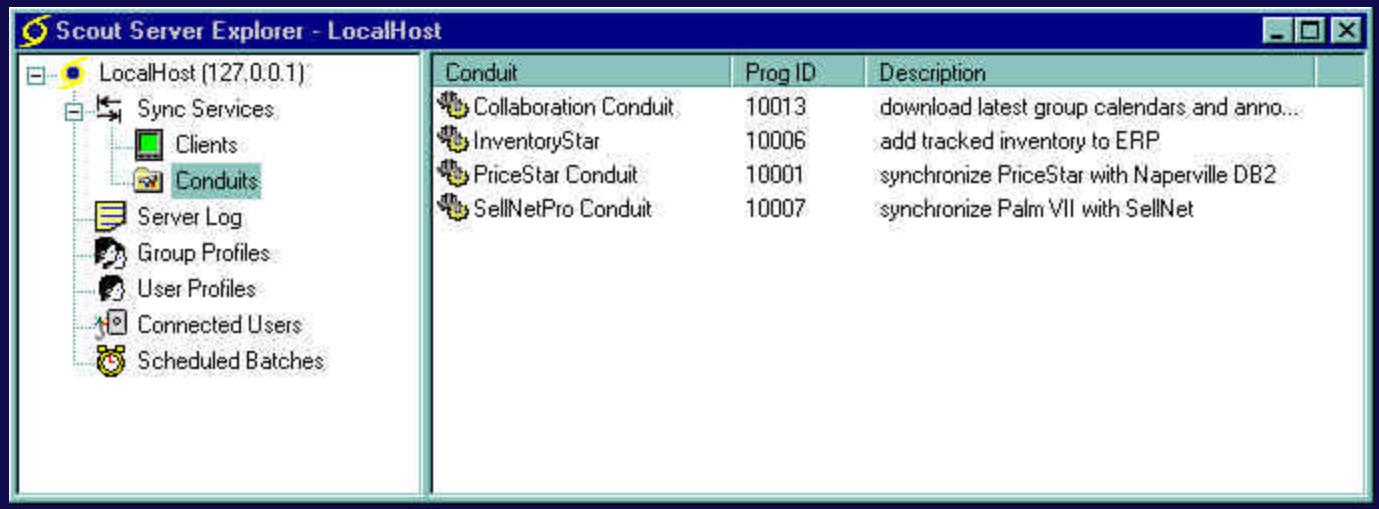
- Sales Force Automation
  - Access data from multiple databases
  - Conduit includes complex business logic



Programmable Conduits

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- ScoutSync (Riverbed Technologies)
  - Hire a programmer





Programmable Conduits

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- Palm HotSync Server
- ScoutSync (Riverbed Technologies)
- WaveSync (WaveWare Communications)

Extend Databases

- DB2 Everywhere (IBM)
- Oracle8i Lite (Oracle)
- UltraLite (Sybase)

Making the Decisions

- Synchronization must fit into the entire system
  - The synchronization architecture
  - The conduit philosophy
  - Product features



Consider Product Features…

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- Connections to your enterprise data
- Administration tools
- Security
- Price and Support

…And Your Infrastructure

PalmSource 99

- Connections to your enterprise data
- Administration tools
- Security
- Price and Support

- Palm application development
- Conduit development
- Training and internal support

Extend the Enterprise

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- A Palm Solution
- Road to the Enterprise
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- What have we learned?




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AvantGo
The Authority in Managing Mobile Information
Rob Meinhardt
Vice President of Product Marketing
3Com
More connected.

AvantGo Background

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- AvantGo Products
  - AvantGo Enterprise 3.1
  - AvantGo.com Interactive Service
- Customers
  - Fortune 500 Enterprise accounts
  - Hundreds of thousands of end-users



- Investment from Both Microsoft and 3Com

Technology Vision

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Any Information

- Web Applications (Internet to Intranet)
- Corporate Data and Enterprise Apps
- Hosted Applications
- Mainstream Content

Any Network

- Desktop Sync
- Direct Dial
- Wireless
- Kiosk

Any Handheld



- 3Com/Palm
- Symbol
- Qualcomm
- Handspring
- HP
- Casio
- Vadem
- Others...

Sample Customers: AvantGo.com & AvantGo Enterprise

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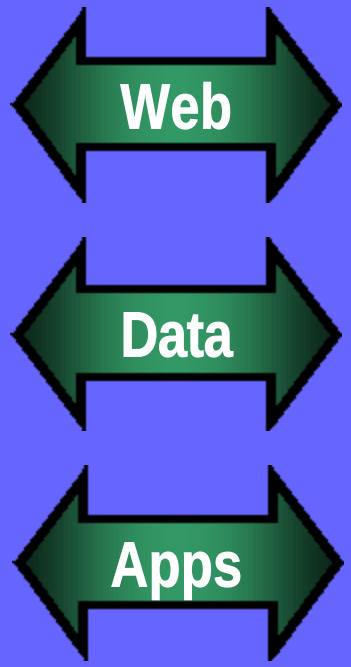
The Complete Enterprise Solution

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Managing the lifecycle…

- ***Develop***
- ***Manage***
- ***Integrate***
- ***Maintain***
- ***Deploy***

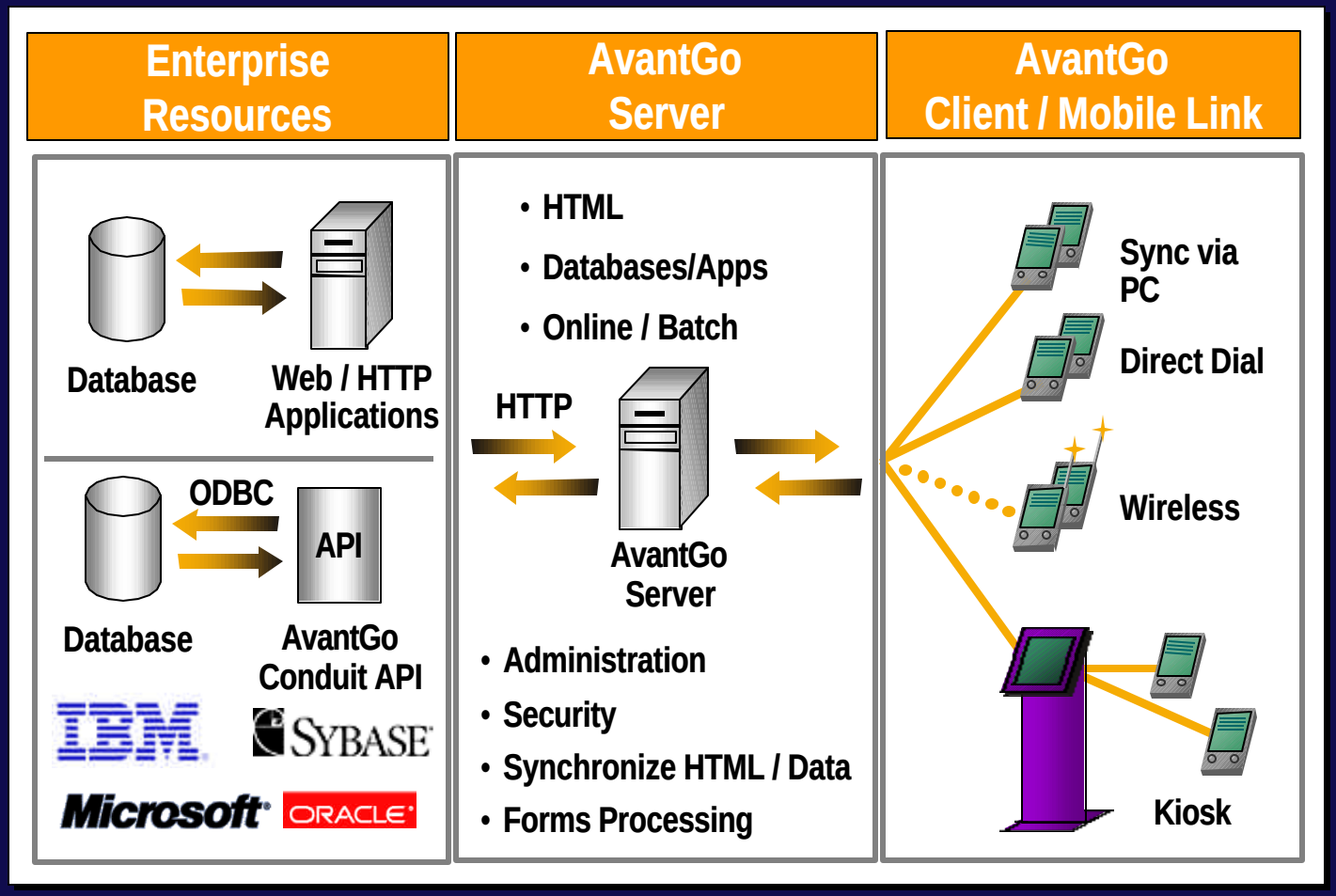
…of your applications





AvantGo…the choice is yours.

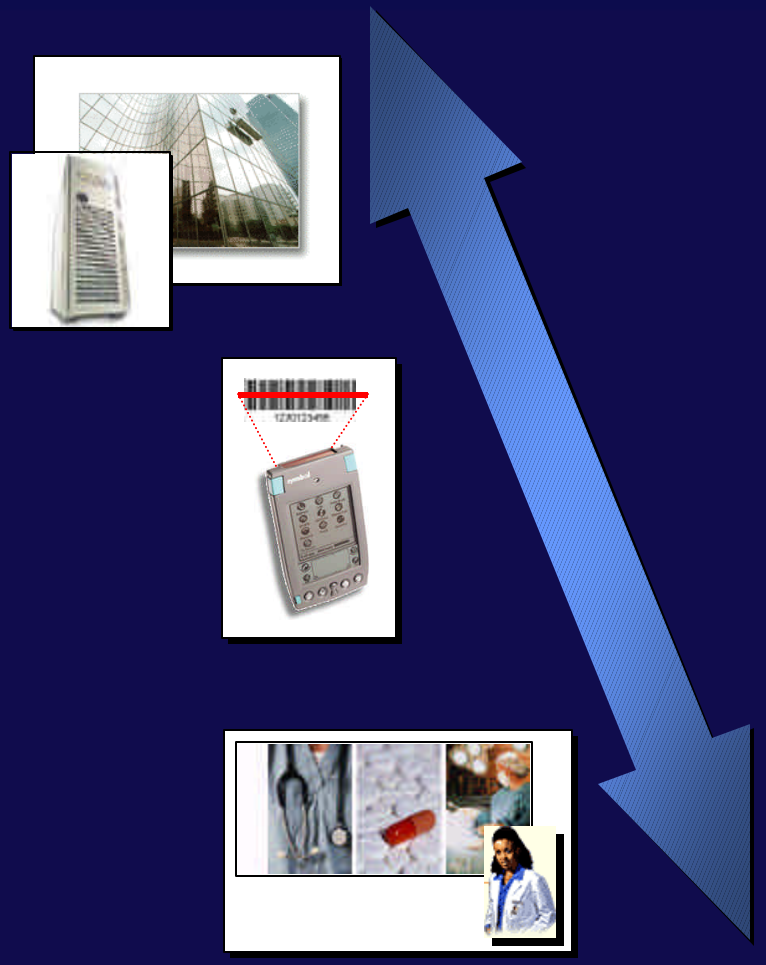
AvantGo Architecture



Supply Chain Healthcare Application

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- Nationwide deployment on Symbol devices
- Healthcare product delivery and billing
- Scan-enabled with signature validated proof of delivery
- Web implementation replaces legacy system
- Multi-million dollar return



Supply Chain Healthcare Application

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- Minimize or eliminate the “Three C’s”
- Web delivers lowest “lifecycle cost”
  - Develop, integrate, deploy and maintain
- End-user performance via dynamic HTML
  - Device as standalone Web server
  - Wireless and disconnected ~ always available
  - High performance interaction
- Systems administration reduces cost
- Scalable and extensible solution


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Riverbed Technologies
The Freedom to Go Mobile
Larry Roshfeld
Vice President
3Com
More connected.

About Riverbed Technologies

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Founded: 1997, Incorporated August 1998

Product Family: ScoutWare

Key Partners: 3Com/Palm, Symbol, Oracle, CA, Aether, OpenSky

Key Customers: MCI, USPS, Becton Dickinson, Sony, AT&T

Focus: Enable a quick, customer responsive and flexible mobile enterprise—extend mission critical applications to mobile and remote workers where they need it, when they need it and in time to take action on it.

ScoutWare

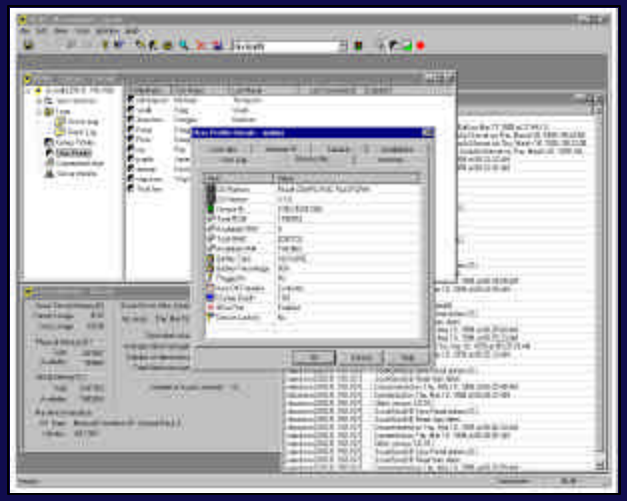
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- **ScoutIT**
  Managing the mobile enterprise
- **ScoutArchitect**
  Designing the mobile enterprise
- **ScoutSync (Palm HotSync Server)**
  Connecting the mobile enterprise
- **ScoutWeb**
  Web enabling the mobile enterprise

ScoutSync (HotSync Server)

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Deliver Client/Server Content to All Mobile Devices

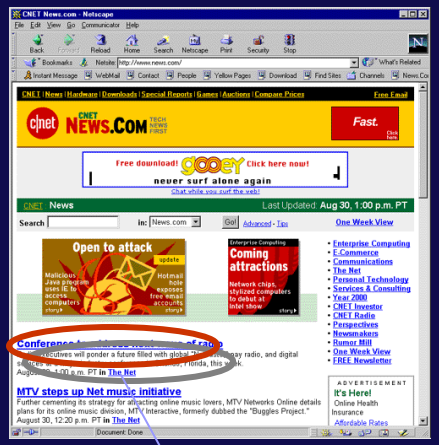


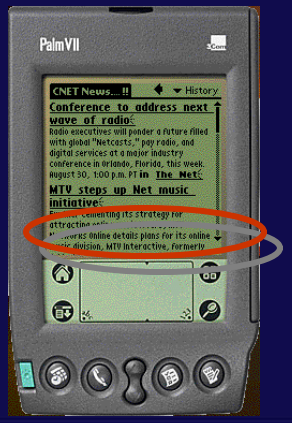
- **Connecting the Mobile Enterprise:** Connect Mobile Workers directly to Enterprise Applications on Network or Host systems
- **Standards Based:** Supports COM, ODBC, VB, VC++, TCP/IP, Palm OS, Windows CE, EPOC
- **Speed Connectivity** to Mission-Critical Business Information via Wired or Wireless connection
- Ensure **Management of Mobile Devices**
- **Lower** Deployment **Costs**
- Effectively and Easily **Manage and Administer** Solution

ScoutWeb

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Deliver Web Content to All Mobile Devices





- **Create Once, Deliver Everywhere Approach:** Eliminates Need to Maintain Multiple Web Content Versions
- **Standards Based:** HTML, XML, XHTML, and WML in any Device Operating System
- **Quickly Leverage Existing Web Applications**—no proprietary software needed on client side
- **Speed Delivery** of Information.
- Ensure **Content Accuracy**
- **Lower** Production **Costs**
- Effectively and Easily **Manage and Administer** Solution

ScoutSync Connects the Mobile Enterprise

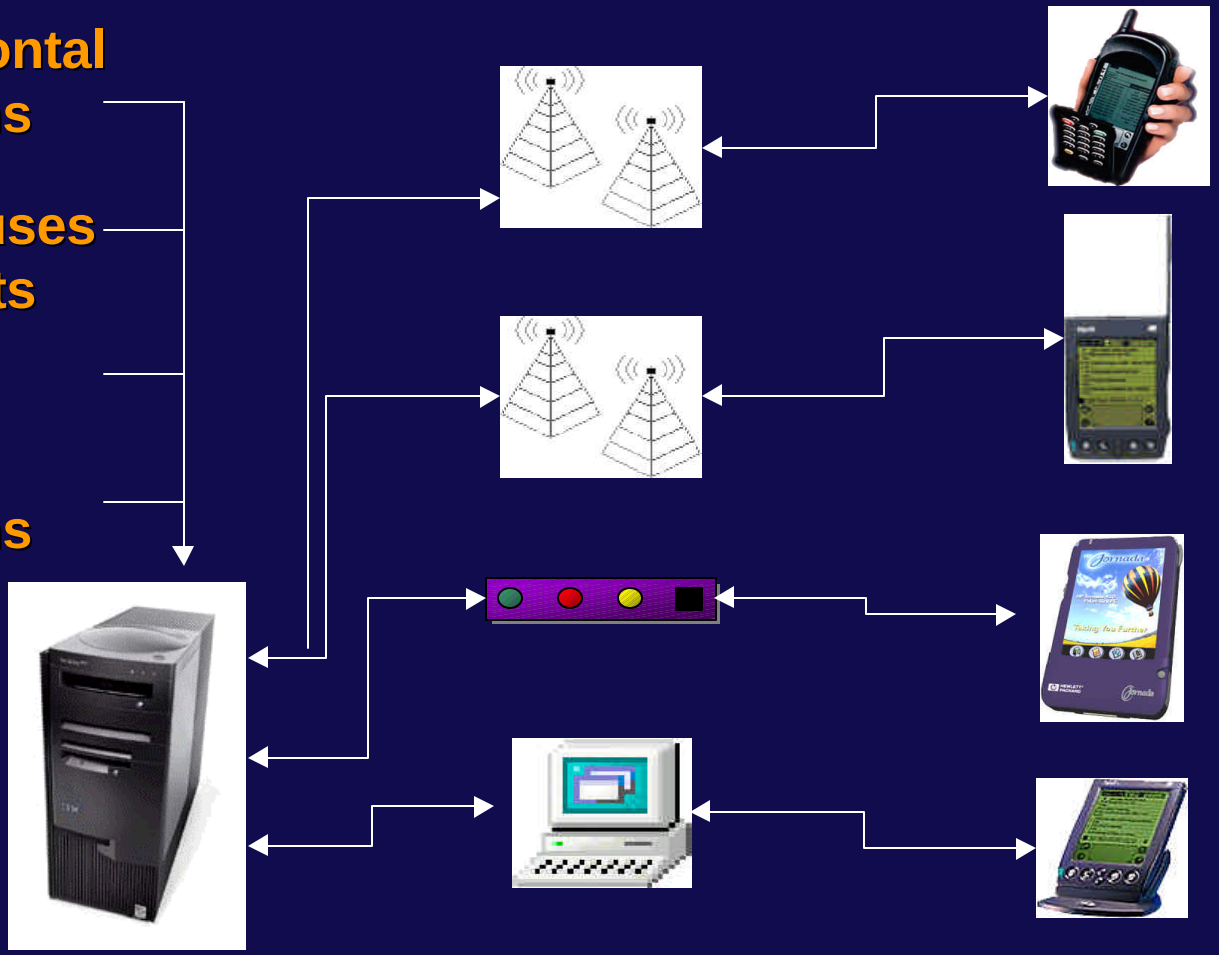
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Vertical/Horizontal Applications

Data Warehouses & Data Marts

ERP Suites

Custom Applications



ScoutSync Enables a Mobile Sales Application

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Mobile Sales Application

- Sales Data
- Market Information
- Business Intelligence
- Mail

Scout Conduits

Scout Server



ScoutSync manages communications and synchronization on mobile clients









Conduits developed using ScoutSync

ScoutSync manages mobile devices

Extend the Enterprise

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- A Palm Solution
- Road to the Enterprise
  - The Handheld
  - The Enterprise
  - Synchronization
- Palm Stories
- What have we learned?



Consider the Entire System

Handheld

Enterprise data



Synchronization

The Menu—Part 1

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Handheld

- Display
- Gather
- Computer
- Combination

Enterprise data

- E-mail / PIM
- Groupware
- Databases
- ERP
- Web based
- Vertical applications

The Menu—Part 2

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Sync Architecture

- Server
- Proxy-Server
- Web-Server

Conduit Type

- Canned
- Managed
- Programmable

Extend the Enterprise

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The Entire System







Handheld **Synchronization** **Enterprise data**

Extend the Enterprise

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Questions







Handheld

Synchronization

Enterprise data


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More connected.